

Problem Set 1 — Speaking Physics

Physics 2026–27 • due at the start of class, Monday, August 31

NAME _____

DATE _____

How to work this set. Show your reasoning, not just answers — on conversions, write the power of ten you used; on estimates, write every guess you multiplied. Honest numbers count here: keep the precision your inputs actually support. This set is formative practice for the **Unit 0 Skills Check on Thursday, September 10**.

1 Name that unit

SI UNITS

For each quantity, give the SI unit we will measure it in *and* its symbol. Then circle the one quantity below whose unit is **derived** (built from base units) rather than a base unit itself.

- a. The mass of a lab cart
- b. The time for the cart to cross the track
- c. The length of the track
- d. The force your hand pushes the cart with

2 Prefixes: back to base

SI UNITS

Convert each measurement to its SI base unit. For each one, follow the four steps from class: identify the prefix, write its power of ten, multiply through, and **sanity-check the direction** (should the number get bigger or smaller?).

- a. A 7.2 km morning run, in meters
- b. A 350 ms human reaction time, in seconds
- c. A 2.4 MW wind turbine, in watts
- d. Green light with a wavelength of 530 nm, in meters (scientific notation is fine here)

WORK

3 Units in a trench coat

SI UNITS

A classmate solves a force problem and proudly reports the answer $F = 12 \text{ m/s}$.

- Without checking a single line of their arithmetic, how do you already know the answer is wrong?
- Write the newton in terms of SI base units, and say what quantity *does* get measured in m/s .

4 Counting significant figures

MEASUREMENT

How many significant figures are in each measurement? For (e), explain in one sentence why the count is ambiguous.

- 0.00470 kg
- 302 m
- $5.60 \times 10^3 \text{ s}$
- 0.5080 L
- 1200 m

5 Read it out loud

VARIABLE NOTATION

Write out each expression in plain English, the way you would read it to the class. Remember: below the line is a *name tag*, above the line is *math*.

- v_i
- x_0
- v_f^2
- F_{net}
- μ_s and μ_k are the same Greek letter wearing two different subscripts. What is the physical difference between them?

6 Match the Greek

GREEK SYMBOLS

Under each symbol, write the letter of its meaning: **(A)** wavelength **(B)** "the change in" **(C)** an angle **(D)** "the sum of" **(E)** angular velocity **(F)** coefficient of friction **(G)** torque **(H)** density

Δ

Σ

θ

ω

λ

μ

ρ

τ

Bonus: one symbol above also moonlights as an SI *prefix*. Which one, and how do you tell which job it's doing?

7 Scientific notation

NOTATION

- Write 92,900,000 m (a satellite's distance) in scientific notation.
- Write 0.0000088 kg (a grain of sand) in scientific notation.
- Write 6.1×10^{-4} s as an ordinary decimal number.
- Which is larger, 3×10^4 or 8×10^3 ? Explain how you can tell *without* writing either number out in full.

8 Fermi problem: the pizza census

ESTIMATION

Estimate: how many slices of pizza do the students of one Miami high school eat in a school year? Do not look anything up — that's the point.

- Break the question into three or four smaller quantities you can reasonably guess, and write down your guess for each one.
- Multiply them together and report your result as an **order of magnitude** (thousands? tens of thousands? hundreds of thousands?).
- Which of your guesses are you least confident in? If it were off by a factor of 2, would your order of magnitude change?

WORK

Graded on the homework 4-3-2-1 scale — completeness and honest reasoning, not just right answers. There is no single correct number for Problem 8; there are only defensible ones.